

VASTRAMS

Vendor & Finance Management System Comprehensive Developer Documentation

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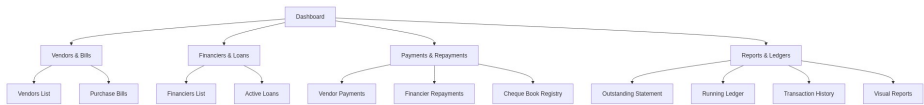
Environment: Replit Hosting (asset-manager Sandbox)

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1. System Overview & Architecture

Vastrams is a double-entry ledger-focused back-office suite designed to resolve accounts payable (Vendors), loan accounting (Financiers), transaction audit trailing, and cash flow forecasting. The application runs on a single-page client architecture powered by React, Tailwind CSS, and a client-side localStorage/mock server engine, providing latency-free transitions and live updates.



1.1 Architecture & State Flow

1. **Routing:** Client-side router handles navigation to twelve distinct views.
2. **Reconciliation Engine:** Utilizes a strict First-In-First-Out (FIFO) algorithm to match payments to older invoices or repayments to older loans automatically.
3. **Auditability:** Implements soft deletes across all transaction tables, allowing items to be marked as deleted while maintaining database integrity.
4. **Data Sync:** Local browser cache syncs back to the sandboxed mock database backend.

2. Detailed Page & Module Specification

2.1 Navigation & Global Layout

The app features a unified structural dashboard containing two major navigation areas:

- **The Sidebar Pane:**
- **Header:** Contains the "Vastrams" logo and sub-brand label "Vendor & Finance".
- **Overview Links:** Dashboard (/).
- **Vendor Module:** Vendors (/vendors), Purchase Bills (/bills).
- **Finance Module:** Financiers (/financiers), Loans (/loans).
- **Payments Module:** Vendor Payments (/vendor-payments), Fin. Repayments (/financier-payments), Cheque Book (/cheques).
- **Reports Module:** Outstanding Statement (/outstanding), Running Ledger (/ledger), Transactions Log (/transactions), Performance Reports (/reports).
- **User Profile (Bottom):** Displays name, role ("Staff"), and the Dark/Light Mode Theme Toggle.
- **The Top Navigation Bar:**
- Displays user profile details, theme switcher status, and a "Sign Out" button.

2.2 Dashboard (/)

The entry point of the app, compiling financial key performance indicators (KPIs) in real-time.

- **Component Cards:**
- **Total Outstanding:** Sum of unpaid vendor bills + active loan balances.
- **Vendor Outstanding:** Subtotal of unpaid purchase bills.
- **Financier Outstanding:** Subtotal of unpaid loan principals.
- **Payments Tracker:** Sum of payments completed today and during the active calendar week.
- **Overdue Alerts:** Displays count and total sum of overdue purchase bills (color-coded red).
- **Cheques Overview:** Displays counts of cheques currently pending.
- **Charts Area:**
- **Outstanding Breakdown:** Comparative bar chart showing vendor liabilities side-by-side with financier loan obligations.
- **Leaderboards:**
- **Top Outstanding Vendors Table:** Displays the top 5 vendors sorted descending by amount. Columns: Code, Vendor Name, Outstanding.
- **Top Outstanding Financiers Table:** Displays the top 5 lenders. Columns: Name, Active Loans, Outstanding.

2.3 Vendors Management (/vendors)

Manages vendor account master files, profiles, and bank settlement details.

- **Vendors Table View:**
- Columns: Code, Vendor Name, Type (e.g. Small/Big Vendor), Outstanding Balance (color-coded green if zero, orange/red if outstanding), Status (Active/Inactive toggle badge), Actions (View, Edit, Delete).
- **Create/Edit Vendor Form (/vendors/new):**
- Vendor Name **Required:** Text input.
- Vendor Type **Required:** Dropdown (Small Vendor, Big Vendor, Fabricator, Weaver, Raw Materials, Service Provider).
- GST Number **Optional:** String validation for 15-character Indian GSTIN format.
- Phone & Address **Optional:** Text fields.
- Bank Details **Optional:**
 - Bank Name: Text input.
 - Account Number: Numeric input.
 - IFSC Code: Text input (validation: 11-char alphanumeric code).
- Item Category **Optional:** Dropdown/Text to categorize goods sourced.
- Opening Balance **Optional:** Positive decimal input (defaults to 0).
- Status **Required:** Dropdown (Active, Inactive).
- Notes **Optional:** Rich text area.

2.4 Purchase Bills (/bills)

Records incoming invoices for goods/services purchased.

- **Purchase Bills Table View:**
- Columns: Bill Number, Vendor Name, Bill Date, Due Date, Bill Amount, Paid Amount, Outstanding Amount, status (Paid, Ongoing, Overdue), Actions (View, Delete).
- **Filters Bar:**
 - Search input for Bill Number.
 - Dropdown to filter by specific Vendor.
 - Dropdown to filter by payment Status (All, Ongoing, Paid, Overdue).
- **Create Purchase Bill Form (/bills/new):**
- Vendor **Required**: Searchable combobox matching active vendors.
- Bill Number **Required**: Unique invoice code.
- Payment Type **Required**: Dropdown (Credit, Cash, Bank Transfer).
- Bill Date **Required**: Date picker (defaults to current date).
- Due Date **Required**: Date picker (validated to be equal to or after Bill Date).
- Amount **Required**: Numeric, must be greater than zero.
- Remarks **Optional**: Text area.

2.5 Financiers Registry (/financiers)

Registry of lenders providing capital financing.

- **Financiers Table View:**
- Columns: Name, Phone, Active Loans Count, Outstanding Loan Balance, Status (Active/Inactive), Actions (View, Edit, Delete).
- **Create/Edit Financier Form (/financiers/new):**
- Name **Required**: Text input.
- Phone **Optional**: 10-digit number.
- Address **Optional**: Text field.
- Status **Required**: Dropdown (Active, Inactive).
- Notes **Optional**: Text area.

2.6 Loans Module (/loans)

Manages loans, tracking principal amounts and interest rates.

- **Loans Table View:**

- Columns: Note Number, Financier Name, Loan Date, Principal Amount, Paid Amount, Outstanding Balance, Interest Rate (%), Status (Active, Settled), Actions (Delete).

Filters Bar:

- Search Note Number.
- Financier dropdown filter.
- Status dropdown filter (All, Active, Paid).
- **Create Loan Form (/loans/new):**
- Financier **Required**: Dropdown of active financiers.
- Note Number **Required**: Unique promissory note reference.
- Loan Date **Required**: Date picker.
- Amount **Required**: Numeric, greater than zero.
- Interest Rate (%) **Optional**: Float value (0-100), tracks annual simple interest rate.
- Remarks **Optional**: Text area.

2.7 Vendor Payments (/vendor-payments)

Registers outflows to vendors.

- **Vendor Payments Table View:**
- Columns: Payment Date, Vendor Name, Paid Amount, Payment Mode, Cheque / Ref Number, Remarks, Actions (Delete - soft deletion).
- **Create Payment Form (/vendor-payments/new):**
- Vendor **Required**: Dropdown of active vendors.
- Payment Date **Required**: Date picker (defaults to today).
- Amount **Required**: Positive decimal value.
- Payment Mode **Required**: Dropdown (Cash, UPI, NEFT, RTGS, Cheque).

Conditional Cheque Fields (visible only if Mode is "Cheque"):

- Cheque Number **Required**: 6-digit number string.
- Cheque Date **Required**: Date picker.
- Cheque Book # **Optional**: Text identifier.
- Bank Name **Required**: Origin bank.
- Remarks **Optional**: Text area.

Allocation Preview Interaction:

- **Preview Allocation Button**: Evaluates outstanding bills for the selected vendor using FIFO. Opens a modal detailing how the payment will apply.
- **Save Button**: Commits payment and updates bills.

2.8 Financier Repayments (/financier-payments)

Registers debt repayments to financiers.

- **Repayments Table View:**
- Columns: Repayment Date, Financier Name, Amount Paid, Payment Mode, Cheque / Ref Number, Remarks, Actions (Delete).
- **Create Repayment Form (/financier-payments/new):**
- Identical structure to Vendor Payments.
- Integrates FIFO Allocation Preview against active loan notes.

2.9 Cheques Registry (/cheques)

Central register to track cheque lifecycles.

- **Cheques Table View:**
- Columns: Cheque Number, Cheque Date, Amount, Party Type (Vendor/Financier), Party Name, Bank, Status (Pending, Cleared, Bounced, Cancelled), Remarks, Actions.
- **Status Quick-Change Actions:**
 - **Clear:** Marks cheque as cleared.
 - **Bounce:** Marks cheque as bounced. Reverts the FIFO payment, raising outstanding bills or loans.
 - **Cancel:** Invalidates the cheque without debit/credit impact.
- **Create/Edit Cheque Form (/cheques/new):**
- Cheque Number **Required:** 6-digit number.
- Cheque Date **Required:** Date picker.
- Amount **Required:** Numeric value.
- Party Type **Required:** Dropdown (Vendor, Financier).
- Party Name **Required:** Dropdown of vendors/financiers based on Party Type.
- Bank Name **Required:** Issuing bank.
- Status **Required:** Dropdown (Pending, Cleared, Bounced, Cancelled).

2.10 Outstanding Report (/outstanding)

Aggregation sheet to track outstanding balances.

- **Summary Metric:** Large card showing Total Outstanding.
- **Outstanding Table:**
- Columns: Name, Party Type (Vendor / Financier), Item Count (unpaid bills or active loans), Total Amount, Paid Amount, Outstanding Balance, Oldest Due Date, Days Overdue.

- **Days Overdue Visuals:** Red highlight if `Days Overdue > 30` to alert staff.
- **Filters:** Dropdown filter (All Parties, Vendors only, Financiers only).

2.11 Running Ledger (`/ledger`)

Generates transaction lists with running balances.

- **Filters:**
 - `Party Selector` (Dropdown of all Vendors and Financiers).
 - `Date Range Picker` (Start Date, End Date).
- **Ledger Table:**
 - Columns: `Date`, `Party Name`, `Ref Number` (Bill No / Loan Note / Payment Ref), `Type` (Bill, Payment, Loan, Repayment), `Description`, `Debit` (Inflow/Liability), `Credit` (Outflow/Asset), `Running Balance`.

2.12 Transaction History (`/transactions`)

Audit trail displaying all transactions.

- **Filters:**
 - Type dropdown (All Types, Bill, Payment, Loan, Repayment).
 - Party dropdown.
 - `Show deleted` Checkbox: Shows soft-deleted records.
 - **Audit Table:**
 - Columns: `Date`, `Type`, `Party`, `Reference`, `Description`, `Amount`, `Status` (Active / Deleted).
- Actions:**
- Delete button (active for normal rows, triggers soft delete).
 - Undo button (visible for soft-deleted rows when "Show deleted" is checked, restores the transaction).

2.13 Reports Panel (`/reports`)

Provides tabular summaries and interactive visual charts.

- **Vendor Outstanding Tab:** Summary cards (`Total Bills`, `Unpaid Bills`, `Overdue Bills`) and a detailed summary table.
- **Financier Outstanding Tab:** Summary cards (`Total Loans`, `Active Loans`, `Closed Loans`) and a financier outstanding table.
- **Payments Tab:** Summary cards displaying vendor payments, financier repayments, and a grand total cash outflow metric.
- **Overdue Tab:** Focuses on overdue invoices.

- **Monthly / Yearly Charts Tabs:**
- Interactive Recharts bar graph plotting payment volume by calendar month or year.

3. Core Accounting Logic

3.1 First-In-First-Out (FIFO) Payment Allocation Algorithm

The FIFO allocation logic resolves payments against outstanding liabilities chronologically:

```
def allocate_payment(party_id, payment_amount):
    # Fetch unpaid/partially paid records ordered by Date ascending
    documents = db.query(
        "SELECT * FROM documents WHERE party_id = ? AND outstanding > 0 ORDER BY date ASC",
        party_id
    )

    remaining_payment = payment_amount
    allocations = []

    for doc in documents:
        if remaining_payment <= 0:
            break

        allocated_amount = min(doc.outstanding, remaining_payment)
        doc.paid += allocated_amount
        doc.outstanding -= allocated_amount
        remaining_payment -= allocated_amount

        allocations.append({
            "doc_id": doc.id,
            "ref_number": doc.ref_number,
            "allocated": allocated_amount,
            "fully_paid": doc.outstanding == 0
        })

        db.save(doc)

    return allocations
```

3.2 Soft-Delete Flow

When a user deletes a transaction:

1. The transaction status changes to "deleted".

2. The balances of all associated documents are recalculated by running the FIFO engine on only "active" transactions.
3. The running balance ledger updates dynamically.